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Optimal Sensor Placement for Structural Health Monitoring of Buildings Using a Kalman Filter-Based Approach

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Abstract

This study proposes a Kalman filter-based method to optimize the placement of accelerometers in buildings, formulated as a multi-objective problem that simultaneously minimizes the number of sensors and the state estimation error. State-space equations of 3-, 9-, 15-, and 30-story buildings were developed from a simplified continuous beam model, allowing the method to be evaluated across different structural conditions. The trace of the state error covariance matrix (Tr(P)) was employed as the performance metric, showing a strong correlation with the signal-to-noise ratio (SNR) and the normalized absolute estimation error. The results highlight that measurement noise critically affects sensor placement. As the noise covariance increases, estimation uncertainty grows, and more sensors are required, often concentrated in specific structural regions. Conversely, high-sensitivity low-noise sensors reduce uncertainty, though at a higher sensor unit cost. Maintaining an SNR above 10 dB proved essential to ensure reliable operational modal analysis. Optimal layouts tended to concentrate on upper floors, where accelerations and SNR are higher, avoiding redundant sensors at modal nodes or lower levels. Validation under real and synthetic excitations, including the 2010 Concepción ground motion record and band-limited white noise, confirmed that the method can accurately identify the fundamental frequencies of the structures. These findings demonstrate the effectiveness of the proposed Kalman filter-based methodology for optimizing sensor layouts in structural health monitoring applications under realistic operational conditions.

Keywords: structural health monitoring; optimal sensor placement; Kalman filter; multi-objective optimization; state estimation; measurement noise; genetic algorithm; Pareto front; accelerometers

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1. Introduction

Structural Health Monitoring (SHM) is a crucial tool to ensure the safety and proper functioning of civil structures. It allows continuous assessment of structural and environmental parameters, facilitating early detection of failures and reducing catastrophic risks [1]. Structures are exposed to gradual or sudden deterioration caused by factors such as intensive use and adverse environmental conditions. This damage, in addition to costly repairs and loss of functionality, represents a significant risk to the lives of users and adjacent structures. Therefore, the development and implementation of SHM systems are essential to improve safety and optimize preventive and corrective maintenance decisions.

In civil engineering, complex structures are susceptible to disturbances that modify their modal characteristics, including overloads that can generate reversible changes of up

to 30%, while environmental variations, such as temperature and humidity, can alter these properties by up to 50%, even without visible damage [2]. These variations reinforce the importance of implementing effective structural health monitoring systems to assess and ensure the safety as well as the functionality of structures over time.

Thus, SHM seeks to answer five key questions over the system in study, as posed by Farrar & Worden [3]: (i) damage presence; (ii) damage location; (iii) damage type; (iv) damage severity; and (v) remaining useful life. These questions guide the design of SHM systems and reflect their value in making informed structural decisions.

To implement SHM in buildings, it is essential to have a network of sensors capable of capturing and translating relevant information about the state of the structure. Moreover, Worden et al. [4] highlight that sensors do not measure damage directly (Axiom IVa), but rather magnitudes such as accelerations or displacements, and that measurements more sensitive to damage are also more sensitive to changing environmental conditions (Axiom IVb). In addition, Axiom VI highlights the trade-off between sensitivity to damage and the ability of an algorithm to reject noise, which represents a key challenge in the design of robust structural monitoring systems.

Recent advances in structural health monitoring have increasingly incorporated data-driven methodologies, including deep learning, deep reinforcement learning, and swarm intelligence-based optimization techniques for damage detection and optimal sensor placement. Comprehensive reviews highlight this shift toward intelligent and adaptive SHM frameworks, particularly in data-rich applications [5]. Recent studies have also demonstrated the practical application of swarm intelligence and hybrid AI models for structural damage identification, showing improved accuracy compared to traditional approaches [6,7]. However, these methods often require large amounts of training data and involve complex model calibration, which may limit their applicability in building monitoring scenarios with sparse instrumentation and limited historical data.

One of the most critical aspects in sensor placement for structural health monitoring is the signal-to-noise ratio (SNR), as it directly affects the accuracy of the extracted modal information and the reliability of the identified parameters. Several studies have demonstrated that low SNR levels increase the uncertainty of modal parameters and may even lead to the loss of identifiable modes [8,9]. Dorvash & Pakzad [10] experimentally showed that low-noise accelerometers provided more consistent and accurate modal parameters than noisier sensors when applied to the Golden Gate Bridge, confirming the direct relationship between noise level and parameter accuracy. Similarly, Ravizza et al. [11] reported that, for low-cost accelerometers, modal identification quality degrades rapidly when the SNR falls below 10 dB. Jahangiri et al. [12] further highlighted that the required SNR threshold also depends on the identification method and the modal order, with the first mode being more sensitive to noise than higher-order modes. In addition, Tan & Zhang [13] emphasized that poor sensor placement can reduce the energy of the measured response, significantly lowering the SNR and complicating modal identification. These findings reinforce the importance of explicitly accounting for SNR in optimal sensor placement strategies.

Although technological advances have enabled the development and use of a greater number of sensors, their distribution is still limited by the high costs of data acquisition systems and accessibility constraints, especially in structures that require monitoring in operating conditions, where sensors are often embedded and not easily relocatable [2]. Therefore, the sensor placement optimization problem has been addressed in the literature by three different classes of algorithms. Convex Relaxation [14] reformulates the combinatorial problem as a convex optimization of the determinant of the estimation error matrix, allowing efficient suboptimal solutions with low computational cost. The Greedy approach [15] selects sensors sequentially by maximizing metrics such as log-determinant

or minimizing reconstruction error, standing out for its computational efficiency and scalability. Finally, metaheuristic (evolutionary) methods have demonstrated a strong ability to explore search spaces in complex engineering problems. Beyond traditional approaches, various metaheuristic algorithms have been successfully applied to the optimal sensor placement (OSP) problem, including simulated annealing [16], the monkey algorithm [17], the ant colony optimization [18], and particle swarm optimization [19]. In this context, genetic algorithms (GA) remain a preferred choice due to their versatility in handling discrete design variables and multi-objective constraints [20].

A promising approach for optimal sensor placement is employing a Kalman filter, which allows the dynamic states of the structure to be estimated from noisy measurements. By minimizing the trace of the covariance matrix of the estimation error, it has proven a feasible evaluation criterion to identify optimal sensor configurations, as demonstrated by its application in sequential schemes [21] and in mixed configurations using the modal Kalman filter under unknown inputs [22]. However, the state estimation can be similarly evaluated by other metrics, such as Fisher entropy, modal energy combined with modal correlation criterion (MSE+MAC), or mutual information, depending on the methodological approach and monitoring objective [23]. In addition, multi-objective methods integrating metrics such as covariance sensitivity matrix and response correlation [24], as well as techniques based on Kriging interpolation to maximize modal information and minimize the number of sensors through Pareto fronts [25] have been proposed. In parallel, recent studies have sought to improve modal coverage in mid-rise buildings by arranging sensors at corners to capture torsional modes [26]. These developments reflect the diversity and evolution of techniques employed to design efficient and robust configurations in SHM systems. In this context, the Kalman filter-based framework is positioned as a complementary and robust alternative for optimal sensor placement, providing a reliable monitoring performance criterion and state estimation under noisy measurements.

Hence, this study seeks to optimize sensor placement on a tall building using a Kalman filter-based approach, addressing a multi-objective problem that aims to minimize, on the one hand, the trace of the state error covariance matrix (i.e., improved accuracy) and, on the other hand, the number of sensors required (i.e., higher efficiency). To solve this problem, we employed a multi-objective genetic algorithm, which allows exploring the sensor configurations and constructing the Pareto front. This methodology is applied to a set of two-dimensional tall building models. Additionally, the results are validated by numerical time-history simulations to evaluate sensor deployment performance.

The remainder of this paper is organized as follows. Section 2 presents the theoretical background, describing the simplified structural model and the Kalman filter formulation used for state estimation. Section 3 details the multi-objective optimization framework, defining the design variables, objective functions, and the multi-objective optimization algorithm employed. Section 4 presents the case studies based on four synthetic building prototypes (3, 9, 15, and 30 stories) for proof-of-concept validation. This section analyzes the optimization results for different building heights and assesses the optimal sensor configurations through time-history simulations involving both synthetic white noise and a real seismic record from the 2010 Maule earthquake (Chile). Finally, Section 5 offers the concluding remarks and suggestions for future research.

2. Methodology

The methodology begins with the modeling of the structures under study, utilizing four synthetic building prototypes (3-, 9-, 15-, and 30-story) to perform a proof-of-concept validation of the proposed framework. These structures are represented through simplified models to generate the corresponding state-space systems for numerical analysis. Ac-

celerometers are then selected as measurement devices, taking into account their sensitivity to noise; a minimum signal-to-noise ratio (SNR) of 10 dB is adopted to ensure reliable modal identification.

The optimization problem is formulated as a multi-objective task, where the Kalman filter is employed to evaluate the performance associated with each sensor configuration. The solutions are synthesized into a Pareto front, considering two competing objectives: minimizing the cost associated with the number of sensors and reducing the trace of the state error covariance matrix (P). To approximate this Pareto front, the gamultiobj() function in MATLAB version R2023a is used to efficiently explore the design space and identify non-dominated solutions. Finally, the optimal sensor layouts are determined and applied to the different building cases, allowing a comparative analysis of the results and the evaluation of simulated sensor measurements under both synthetic and real seismic excitations.

2.1. Simplified Model

The simplified model proposed by Miranda & Taghavi [27] provides an approximate method for calculating the ground acceleration demands at any level of a multi-story building, assuming an elastic or nearly elastic response during earthquake motions. The method employs a continuous structure model combining a flexural beam and a shear beam connected by infinitely rigid axial links to represent the building's dynamic characteristics. The procedure relies on three non-dimensional parameters (α_0, δ, λ) which physically describe the deformation behavior and stiffness distribution of the structure.

The dimensionless parameter α_0 governs the type of lateral deformation behavior. Physically, it represents the ratio of flexural stiffness to shear stiffness. A value of $\alpha_0 = 0$ corresponds to a purely flexural cantilever model (e.g., a shear wall building), while a value of $\alpha_0 \to \infty$ corresponds to a purely shear model (e.g., a moment frame building). Intermediate values of α_0 describe dual systems. In practical terms, buildings with shear walls or braced frames usually have values between $0 \leq \alpha_0 \leq 1.5$; while buildings with moment-resisting frames usually have values $5 \leq \alpha_0 \leq 20$ [27].

To describe how the lateral stiffness varies along the building height, the model uses the following nondimensional shape equation:

$$k(x) = 1 - (1 - \delta)x^{\lambda} \tag{1}$$

where x is the normalized height (0 for ground level, and 1 for roof level). The parameter δ represents the stiffness taper ratio, defined as the ratio of the lateral stiffness at the top story to that at the first story. Physically, a $\delta < 1$ accounts for the common practice of reducing structural sections (columns and walls) at higher floors; $\delta = 1$ would imply a uniform stiffness distribution along all floors. Finally, the parameter λ is the exponent of the stiffness variation, which dictates the rate at which stiffness reduces with height. A value of $\lambda = 1$ represents a linear reduction, while $\lambda = 2$ represents a parabolic (quadratic) reduction, allowing the model to fit various architectural mass and stiffness distributions.

By defining non-dimensional parameters (α_0, δ, λ) along with the fundamental natural period (T_1), the simplified model allows for the derivation of closed-form solutions for approximated dynamic properties for higher modes "r" such as mode shape vectors (ϕ_r), period ratios (T_r/T_1), and participation factors (Γ_r), which are requisite inputs for generating the state-space model as described in the following section. All the specific details and mathematical formulas derived from this model are available in Miranda & Taghavi [27].

Regarding the fundamental period (T_1), we employed the empirical equations by Guendelman et al. [28], based on the statistical analysis of structural and dynamical properties of Chilean tall buildings, including 4105 reinforced concrete buildings constructed between

1993 and 2017, which allowed generating a robust classification to evaluate the dynamic behavior of a wide range of buildings against severe seismic events. The authors proposed a building classification based on the H_b/T_1 ratio, where H_b is the total roof height of the building. Three building categories are described: (i) flexible (20 [m/s] < H_b/T_1 < 40 [m/s]); (ii) normal (40 [m/s] < H_b/T_1 < 80 [m/s]); and (iii) rigid (80 [m/s] < H_b/T_1 < 150 [m/s]). Buildings with H_b/T_1 values below 20 [m/s] or above 150 [m/s] fall outside typical ranges, indicating outliers with extreme flexibility or stiffness.

2.2. State Space Model

The state-space model is a mathematical framework that describes the time evolution of a dynamic system through equations that relate its internal states, inputs, and outputs [29]. Figure 1 presents a block diagram illustrating this process, where the system is driven by two distinct inputs: the known control input $u(t)$ and the disturbance $w(t)$, which models external uncertainties. The State Space Model block simulates the building's dynamic behavior based on these inputs and its structural characteristics, generating a theoretical output $y_t(t)$ that represents the ideal, noiseless response. Finally, measurement noise $v(t)$ is added to this theoretical signal to produce the final observed output $y(t)$, effectively capturing the reality of sensor data collection.

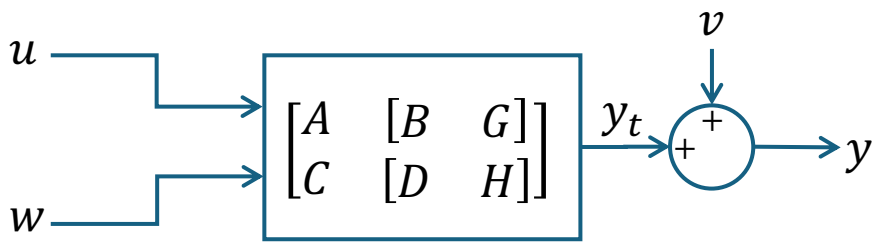


Figure 1. Diagram of the inputs and outputs of the State-Space Model.

The explanation of how the state-space model is generated begins with the equation of motion. In this case, Equation (2) presents the equation of dynamic motion (EDM) in modal coordinates associated with the "r" mode:

$$\ddot{q}_r(t) + 2\xi_r\omega_r\dot{q}_r(t) + {\omega_r}^2 q_r(t) = \Gamma_r\, \ddot{U}_g(t) \tag{2}$$

where $\ddot{q}_r$, $\dot{q}_r$ and q_r are the acceleration, velocity, and displacement of mode "r", respectively; ω_r y ξ_r are the natural frequency and the damping ratio of the mode "r"; Γ_r is the modal participation factor of the external input of mode "r"; and $\ddot{U}_g(t)$ represents an exogenous input to the system, which in this study corresponds to ground acceleration due to an earthquake.

This equation of motion (which consists of a second-order ordinary differential equation) can be reformulated and simplified as a first-order equation in terms of state variables, which allows representing the dynamics of the system in a more manageable format for analysis and control, taking the following form:

$$\dot{x}(t) = Ax(t) + Bu(t) + Gw(t) \tag{3}$$

where $x(t)$ represents the state vector, which describes the internal variables of the system; A models the evolution of the system's states over time; B defines how known entries $u(t)$ affect the system; and G defines how the disturbance $w(t)$ affects the dynamics of the system.

Measurements of the dynamic response of the structure y are expressed as linear functions of the states, incorporating measurement noise v, according to the following equation:

$$y(t) = Cx(t) + Du(t) + Hw(t) + v(t) \tag{4}$$

where C links internal states to the observable outputs; D directly relates the known entries $u(t)$ with the measurements; and H describes how the disturbance $w(t)$ affects measurements.

Then, to generate the matrices A, B, G, C, D, and H to form the first-order equations of the state-space model, the following is considered:

$$A = \begin{bmatrix} 0_{n\times n} & I_{n\times n} \\ -\text{diag}(\omega_r^2) & -\text{diag}(2\xi_r\omega_r) \end{bmatrix}_{2n\times 2n} \tag{5A}$$

$$B = G = \begin{bmatrix} 0_{n\times 1} \\ \{\Gamma_r\}_{n\times 1} \end{bmatrix}_{2n\times 1} \tag{5B}$$

$$C = \begin{bmatrix} -\Phi\text{diag}(\omega_r^2) & -\Phi\text{diag}(2\xi_r\omega_r) \end{bmatrix}_{n\times 2n} \tag{5C}$$

$$D = H = 0_{n\times 1} \tag{5D}$$

where $\text{diag}(\cdot)$ is the "$n \times n$" diagonal matrix, with diagonal entries as stated in its corresponding argument; $\{\Gamma_r\}_{n\times 1}$ is the "$n \times 1$" vector that collects all modal participation factors; and $\Phi = [\phi_1, \phi_2, \ldots, \phi_n]$ is the "$n \times n$" mode shape matrix.

In the development of the model, "n" is considered to represent the number of degrees of freedom of the structure, which defines the dimension of the state vector. The disturbance $w(t)$ is assumed to affect the system in the same way as the known input $u(t)$. Therefore, in this study, it is considered that the matrices G and B are equal, i.e., $G = B$.

On the other hand, the matrix C is constructed so that the measurements correspond to the absolute acceleration of each floor. Furthermore, when sensors are not installed on all floors of the building, the system response matrices are adjusted by keeping only the rows corresponding to the locations where sensors are installed. This modification is reflected in the C, D and H, ensuring that the model outputs correspond only to the available measurements. More information is presented in Section 3.

2.3. Signal-to-Noise Ratio (SNR)

The signal-to-noise ratio (SNR) is a fundamental metric in measurement systems that evaluates the quality of the signal obtained relative to the noise present.

$$\text{SNR} = 10\log_{10}\left[\left(\frac{A_{\text{signal}}}{A_{\text{noise}}}\right)^2\right] = 20\log_{10}\left(\frac{A_{\text{signal}}}{A_{\text{noise}}}\right) = A_{\text{signal,dB}} - A_{\text{noise,dB}} \tag{6}$$

where A_{signal} is the RMS (Root Mean Square) value of the useful signal, which represents the effective magnitude of the structural response; A_{noise} is the RMS value of the noise in the measurement; $A_{\text{signal,dB}}$ and $A_{\text{noise,dB}}$ represents the signal and noise levels in decibels (dB).

The RMS value of the signal is defined as

$$A_{\text{signal}} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\text{signal}[k]^2} \tag{7}$$

where N is the sample size. A similar definition is valid for A_{noise}.

The RMS value (A_{signal}) provides a measure of the energy contained in the signal, making it a useful tool for evaluating both signals and noise in practical applications. In

terms of structural engineering, a high SNR indicates that the dynamic responses of the structure are more representative, while a low SNR suggests that noise significantly affects the measurement quality [30].

In this research, the sensors are assumed to measure absolute acceleration. To assess sensor quality, the signal-to-noise ratio (SNR) is expressed in decibels, and only acceleration sensors with SNR values higher than 10 dB are considered, ensuring a minimum level of measurement reliability. The adoption of this 10 dB threshold is supported by previous studies that demonstrated the sensitivity of modal parameters to measurement noise. Ravizza et al. [11] reported that, when working with low-cost sensors, the quality of modal identification degrades rapidly when the SNR falls below 10 dB. Similarly, Dorvash & Pakzad [10] experimentally showed, using data from the Golden Gate Bridge, that sensors with lower noise levels provide more consistent and accurate modal parameter estimates than noisier sensors. Furthermore, Jahangiri et al. [12] highlighted that the required SNR threshold also depends on the identification method: for beam structures, the first mode requires SNR $\geq$ 13.98 dB using Peak Picking method, while higher modes can be identified with lower values. Considering these findings, a minimum threshold of 10 dB is established in this work as a representative and conservative value to ensure the reliability of the estimated modal parameters.

2.4. Disturbance and Measurement Noise

The disturbance $w(t)$ and measurement noise $\nu(t)$ represent unpredictable phenomena, such as model uncertainty and sensor inaccuracies, respectively. Both are modeled as stationary Gaussian white noise processes with zero mean:

$$\mathbb{E}[w(t)] = 0, \quad \mathbb{E}[\nu(t)] = 0 \tag{8}$$

where $\mathbb{E}[\cdot]$ is the expectation operator. Their corresponding covariance matrices, Q_w and R_v, are defined as:

$$\mathbb{E}[w(t)w^T(\tau)] = Q_w\delta(t-\tau) \tag{9}$$

$$\mathbb{E}[\nu(t)\nu^T(\tau)] = R_v\delta(t-\tau) \tag{10}$$

where $\delta(\cdot)$ is the Dirac delta function. These equations indicate that the processes are uncorrelated in time. Therefore, the covariance can be written in a simplified form:

$$\mathbb{E}[w(t)w^T(t)] = Q_w, \quad \mathbb{E}[\nu(t)\nu^T(t)] = R_v \tag{11}$$

To characterize the relationship between these noise sources, a proportionality factor $\beta \geq 1.0$ is introduced:

$$R_v = \beta Q_w \tag{12}$$

In this study, the process noise covariance Q_w was assumed to be significantly lower than the measurement noise covariance R_v, with Q_w set to a constant value close to zero ($0.001\ [\mathrm{m}^2/\mathrm{s}^4]$). This assumption reflects a scenario where the structural model is considered reasonably accurate and the structure operates strictly within the linear elastic range. By minimizing the uncertainty attributed to modeling errors (e.g., non-linearities or degradation), this approach allows for the isolation of sensor noise effects, thereby enabling a focused evaluation of how sensor placement and quality (R_v) influence state estimation performance.

2.5. Kalman Filter

The Kalman filter is a computationally efficient algorithm used to estimate the internal states of linear dynamic systems subject to unmeasured disturbances and measurement

noise [31]. As illustrated in Figure 2, the filter operates recursively to predict the system state and minimize the associated estimation uncertainty

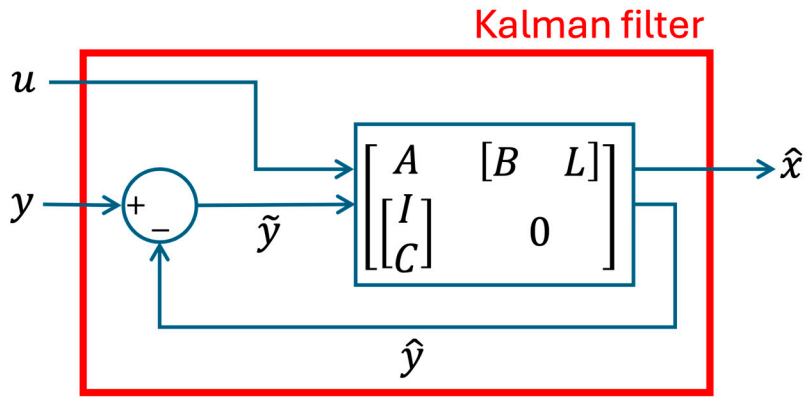


Figure 2. Kalman filter block diagram.

The primary objective is to minimize the deviation between the actual and estimated states, expressed as

$$\text{Minimize } \tilde{y} = y - \hat{y} \iff \text{Minimize } \tilde{x} = x - \hat{x} \tag{13}$$

where $\tilde{y}$ is the output estimation error; $\tilde{x}$ the state estimation error; $\hat{y}$ is the output estimation; and $\hat{x}$ the state estimation. To achieve this, the filter computes the error covariance matrix P, a symmetric positive-definite matrix that quantifies the estimation uncertainty. This matrix is obtained by solving the continuous algebraic Riccati equation:

$$AP + PA^T + GQ_wG^T - PC^T{R_v}^{-1}CP = 0 \tag{14}$$

Once P is determined, the Kalman gain L is computed:

$$L = PC^T{R_v}^{-1} \tag{15}$$

Finally, the state estimate and system output are updated using the computed gain:

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - \hat{y}) \tag{16}$$

$$\hat{y} = C\hat{x} \tag{17}$$

In this study, the MATLAB function kalman() was employed to automate the solution of the Riccati equation and the computation of the gain L and state estimator, utilizing the system matrices (A, C) and noise covariances (Q_w,R_v).

To ensure stability and proper performance of the filter, the following conditions must be met: (i) (C, A) must be observable, which is verified if the observability matrix has full rank, that is, its rank is equal to the number of states of the system; (ii) matrix A must be Hurwitz, meaning all its eigenvalues have negative real parts; and (iii) the states must be detectable, which implies that any unobservable state is associated with stable modes of the system.

However, it is important to mention that the classical concept of observability does not consider the inherent noise in the measurements. In systems with measurement noise, such as those addressed by the Kalman filter, the concept of "stochastic observability" is introduced. This approach evaluates whether the filter can keep the error covariance matrix bounded in the presence of noise, ensuring that the estimates do not grow indefinitely. A relevant methodology is the Stochastic Observability Test proposed by Bageshwar et al. [32],

which verifies this condition using singular values of matrices related to the system, providing a more robust design under uncertainty.

3. Multi-Objective Optimization Problem

This section details the formulation of the optimal sensor placement (OSP) strategy as a multi-objective optimization problem. The approach aims to balance two conflicting goals: reducing the instrumentation cost by minimizing the number of sensors and maximizing the accuracy of the structural state estimation. To this end, the design variables representing potential sensor locations are first defined. Subsequently, the objective functions are established, integrating the Kalman filter-based performance metric. Finally, the optimization framework is formally stated to generate the Pareto front, which identifies the set of non-dominated solutions representing the most efficient trade-offs for the monitoring system.

3.1. Design Variable

The design variable corresponds to a Boolean vector, defined as

$$S = [s_1 \ \dots \ s_n \,], \quad s_i \in \{0,1\}, \ \forall i \in \{1,\ 2,\ \dots,\ n\} \tag{18}$$

where $s_i = 1$ indicates that there is a sensor in the floor "i"; $s_i = 0$ indicates that no sensor is placed on floor "i"; "n" is the total number of floors of the structure; and "m" is the number of sensors used.

3.2. Objective Functions

The first objective function f_1 seeks to minimize the cost associated with the number of sensors used:

$$f_1{}^*(S) = \frac{f_1(S)}{f_1{}^{\max}} \quad , \qquad f_1(S) = \sum_{i=1}^{n} s_i = m \tag{19}$$

where $f_1{}^*(S)$ corresponds to the standardization of $f_1(S)$ with respect to the maximum value $f_1{}^{\max}$.

On the other hand, the second objective function f_2 seeks to minimize the trace value of the state error covariance matrix (Tr(P)), which is obtained from the Kalman filter considering the location of the sensors S.

$$f_2{}^*(S) = \frac{f_2(S)}{f_2{}^{\max}} \quad , \qquad f_2(S) = \mathrm{Tr}(P(S)) \tag{20}$$

where $f_2{}^{\max}$ corresponds to the maximum value of $\mathrm{Tr}(P(S))$ evaluated across the combinations provided by the Pareto front.

3.3. Multi-Objective Optimization Problem Definition

The optimization problem is mathematically defined as follows:

$$\min_S \{\alpha f_1^*(S) + (1-\alpha) f_2^*(S)\} \tag{21}$$

subject to n being the number of floors of the structure; $s_i \in \{0,1\}$, $\forall i \in \{1,\ 2,\ \dots,\ n\}$; $\alpha \in \{0,1\}$. The parameter α represents the weight of each objective; for example, $\alpha = 1$ prioritizes $f_1{}^*$ and $\alpha = 0$ prioritizes $f_2{}^*$. Intermediate values balance both objectives.

To solve this combinatorial problem, the concept of the Pareto front is employed. The Pareto front is a graphical tool used in multi-objective optimization that visualizes non-dominated solutions—those in which one objective cannot be improved without worsening

another. For example, Mello et al. [25] present a Pareto front illustrating the relationship between the number of sensors and interpolation error, where each point represents an optimal configuration according to specific priorities.

In this research, the Pareto front is computed using the gamultiobj() solver from the MATLAB Global Optimization Toolbox, which implements a controlled elitist genetic algorithm (NSGA-II). To address the discrete nature of the sensor placement problem, integer constraints were applied using the intcon option. The algorithm was configured with a population size of 200 individuals and a maximum of 200 generations, ensuring robust convergence and adequate exploration of the solution space to identify the set of optimal configurations.

3.4. Optimal Sensor Placement Criteria

To determine the optimal sensor placement, the information presented in the Pareto front is analyzed, which displays the optimized results for each number of sensors. In this graph, the x-axis shows the total cost associated with each optimized sensor configuration (objective function 1), while the y-axis represents the value of $\mathrm{Tr}(P)$ of each sensor configuration (objective function 2), assuming a unit cost per sensor (i.e., a cost of 1 equals one sensor). In addition, the parameter $\alpha = 0.5$ is considered. From this, the optimal solution closest to the origin is selected (minimum distance, Equation 22), reflecting an efficient balance between minimizing the number of sensors and reducing the estimation error.

$$\text{distance} = \sqrt{{f_1}^*(S)^2 + {f_2}^*(S)^2} \tag{22}$$

The overall workflow adopted for this optimization process is schematically illustrated in the flow diagram in Figure 3. It is important to emphasize that this optimization relies on the steady-state error covariance P, which is a function of the system matrices (A, C) and noise statistics (Q_w, R_v). Consequently, the determined optimal sensor configuration is intrinsic to the structural properties and is independent of the specific external loading (u) or earthquake records applied during the subsequent performance simulations.

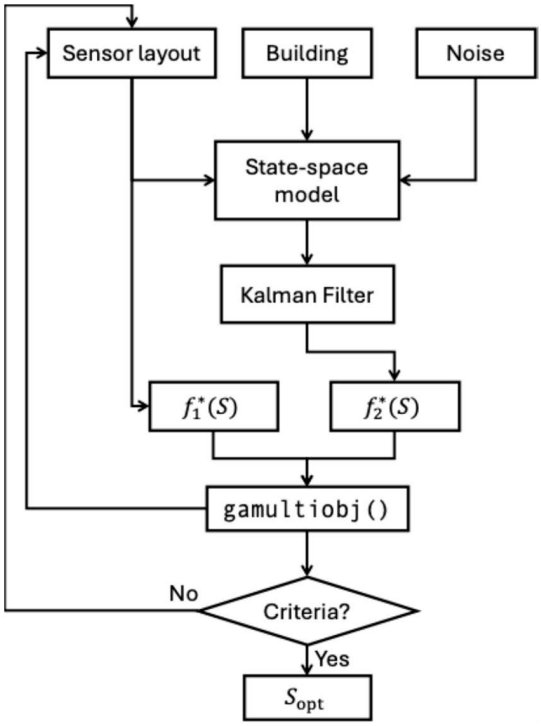


Figure 3. Flow diagram of the sensor optimization process.

3.5. Simulation of Sensor Layout Performance

Once the optimal sensor locations are determined, simulations are conducted in Simulink to generate sensor measurements. These simulations allow for the analysis of state estimations and the identification of the structure's fundamental frequencies. The known input (u) consists of two distinct ground motions used to assess performance. First, the seismic record of the 2010 Maule earthquake (Chile), recorded at the Concepción station in the horizontal direction (Figure 4), is employed to represent the Maximum Considered Earthquake (MCE). This allows for the evaluation of the building's dynamic response under a major seismic event. Second, a band-limited white noise excitation is utilized to simulate a serviceability case, characterized by a flat spectrum between 0 and 100 Hz, followed by a rapid drop in energy, confirming the correct application of a 100th-order low-pass filter and uniform energy distribution within the modal range of interest.

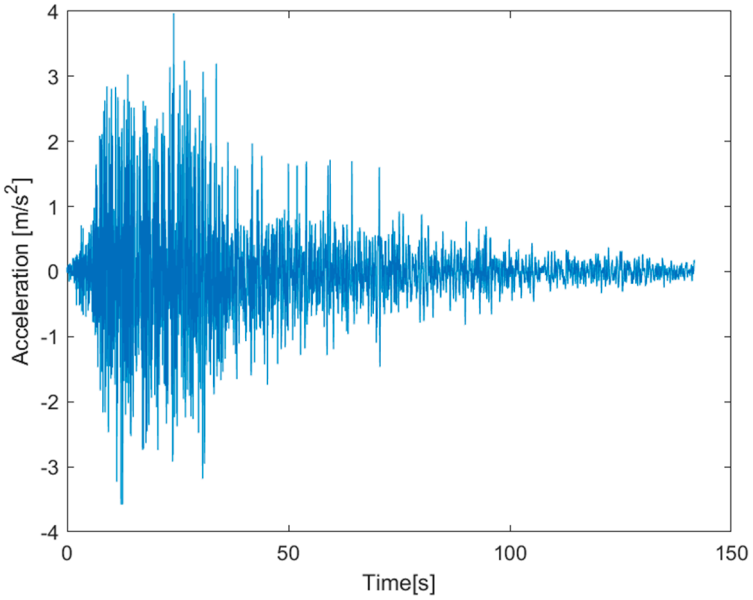


Figure 4. Acceleration record of the 2010 Maule earthquake, Concepción station, channel 3.

Figure 5 displays the Power Spectral Density (PSD) of the 2010 Concepción ground motion. It can be observed that the majority of the energy is concentrated at low frequencies, between 0 and 10 Hz, which is consistent with the expected dynamic behavior of civil structures. This spectral content confirms that the record is sufficient to excite the fundamental vibration modes of the studied buildings.

To assess the accuracy of the state estimation, the Normalized Root Mean Square Error (NRMSE) is employed. This index evaluates the deviation between the estimated and actual responses, normalized with respect to the magnitude of the real signal. The NRMSE is obtained as follows:

$$\mathrm{NRMSE}_{(i)} = \frac{\sqrt{\frac{1}{N}\sum_{k=1}^{N}\left(x_{\mathrm{estimated}(i)}[k] - x_{\mathrm{true}(i)}[k]\right)^2}}{\sqrt{\frac{1}{N}\sum_{k=1}^{N}\left(x_{\mathrm{true}(i)}[k]\right)^2}} \times 100\% \tag{23}$$

where $x_{\mathrm{true}(i)}$ and $x_{\mathrm{estimated}(i)}$ corresponds to the true and estimated values of state "i", respectively. This metric is widely used in structural health monitoring to objectively quantify the accuracy of state estimation models [33].

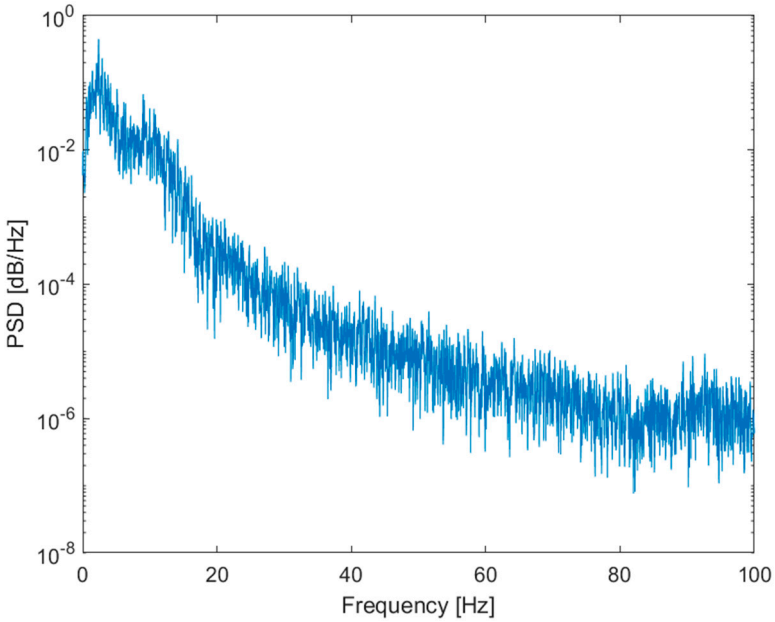


Figure 5. Power spectral density (PSD) of the 2010 Concepción ground motion record.

Finally, a PSD analysis is conducted on the measured responses obtained from sensors positioned according to the different location scenarios. The auto-PSD of these acceleration responses is computed to estimate the natural frequencies of the structure. This analysis serves to assess whether the determined optimal sensor location can accurately estimate these dynamic properties, thereby validating the effectiveness of the proposed configuration for reliable modal identification [34].

4. Results

4.1. Prototype Case Studies

This section outlines the geometric and dynamic properties of the four synthetic building prototypes (3, 9, 15, and 30 stories) utilized for the proof-of-concept validation of the proposed methodology. These models serve as a platform for numerically validating the framework under controlled conditions, ensuring an objective assessment of optimal sensor placement and state estimation performance across diverse structural configurations. Table 1 summarizes the key parameters for each structure, including fundamental periods, frequencies, and the non-dimensional coefficients from the simplified model that define the stiffness distribution. For the dynamic analysis, a constant modal damping ratio of 2% ($\xi_r = 0.02$) was applied to all vibration modes across all building models. Additionally, the corresponding modal shapes are detailed in Appendix A, providing a comprehensive view of the displacement distributions and dynamic behavior of each structure.

Table 1. Parameters for modeling the four building cases under study.

Building	Stories	H_b [m]	Classification (H_b/T_1)	Simplified Model			Period [s]			Frequency [Hz]		
				α_0	δ	λ	T_1	T_2	T_3	f_1	f_2	f_3
1	3	12	Normal	1	0.25	1	0.218	0.042	0.015	4.583	23.782	64.727
2	9	36	Normal	5	0.25	1	0.655	0.207	0.096	1.528	4.837	10.466
3	15	60	Flexible	15	0.25	1	2.400	0.878	0.506	0.417	1.139	1.978
4	30	120	Flexible	15	0.25	1	4.800	1.739	0.994	0.208	0.575	1.006

4.2. Signal-to-Noise Ratio (SNR) Analysis

This section presents the results of the signal-to-noise ratio (SNR) analysis for the considered buildings (3-, 9-, 15-, and 30-story), subjected to a service-level earthquake simulated by white noise with a PGA of 0.2 g. Three noise scenarios were evaluated, as defined in Table 2, corresponding to Cases (A), (B), and (C). In each case, the minimum and maximum SNR values across the floors are reported. The results show that increasing the measurement noise covariance (R_v) produces a significant reduction in SNR values, directly affecting the reliability of modal identification. In Case (A), SNR values remain above 14 dB in all buildings, ensuring the reliability of the extracted parameters. In Case (B), the values decrease to around 4 dB in taller buildings, so only in the 3- and 9-story buildings can the sensors be reliably used, as they maintain an SNR above the 10 dB threshold. Finally, in Case (C), the values fall below this threshold, even reaching negative values in the 15- and 30-story buildings, which indicates that measurement noise surpasses the structural response, severely compromising the ability to obtain reliable modal parameters. For this reason, Case (C) will not be considered in the subsequent analyses due to the elevated noise level.

Table 2. Covariances of process noise and measurement noise for Cases (A), (B), and (C).

	(A)	(B)	(C)	Units
Q_w	0.001	0.001	0.001	$[m^2/s^4]$
β	10	100	1000	-
R_v	0.01	0.1	1	$[m^2/s^4]$

The summary of these results is presented in Table 3, which reports the minimum and maximum SNR values for each building and noise case under the considered excitation.

Table 3. Range of SNR for 3-, 9-, 15-, and 30-story buildings in noise cases (A), (B), and (C), under the excitation of a service-level earthquake modeled as white noise with PGA = 0.2 g.

		Case (A)		Case (B)		Case (C)	
Building	**Stories**	**Min (SNR)**	**Max (SNR)**	**Min (SNR)**	**Max (SNR)**	**Min (SNR)**	**Max (SNR)**
1	3	24.7	26.4	14.7	16.4	4.7	6.4
2	9	21.8	23.0	11.8	13.0	1.8	3.0
3	15	14.9	16.7	4.9	6.7	−5.1	−3.3
4	30	14.2	16.0	4.2	6.0	−5.8	−4.0

4.3. Optimal Sensor Placement: Case Study Results

The following results were obtained by applying the Kalman filter-based method to optimize the placement of sensors (accelerometers) in the case studies (Section 3.1). The analysis considers the process noise covariance Q_w and the measurement noise covariance R_v, as presented in Table 2. In addition, a uniform weighting between both objectives was established by setting $\alpha = 0.5$. For each case, the corresponding Pareto front was constructed, and the optimal solution was selected as the configuration with the shortest distance to the origin, representing the best trade-off between the number of sensors and the estimation error.

Figure 6 presents the Pareto fronts obtained for the four analyzed buildings (3-, 9-, 15-, and 30-story), considering three levels of measurement noise corresponding to Cases

(A), (B), and (C), with β = 10, β = 100, and β = 1000, respectively. In each plot, the curves illustrate the trade-off between the two optimization objectives.

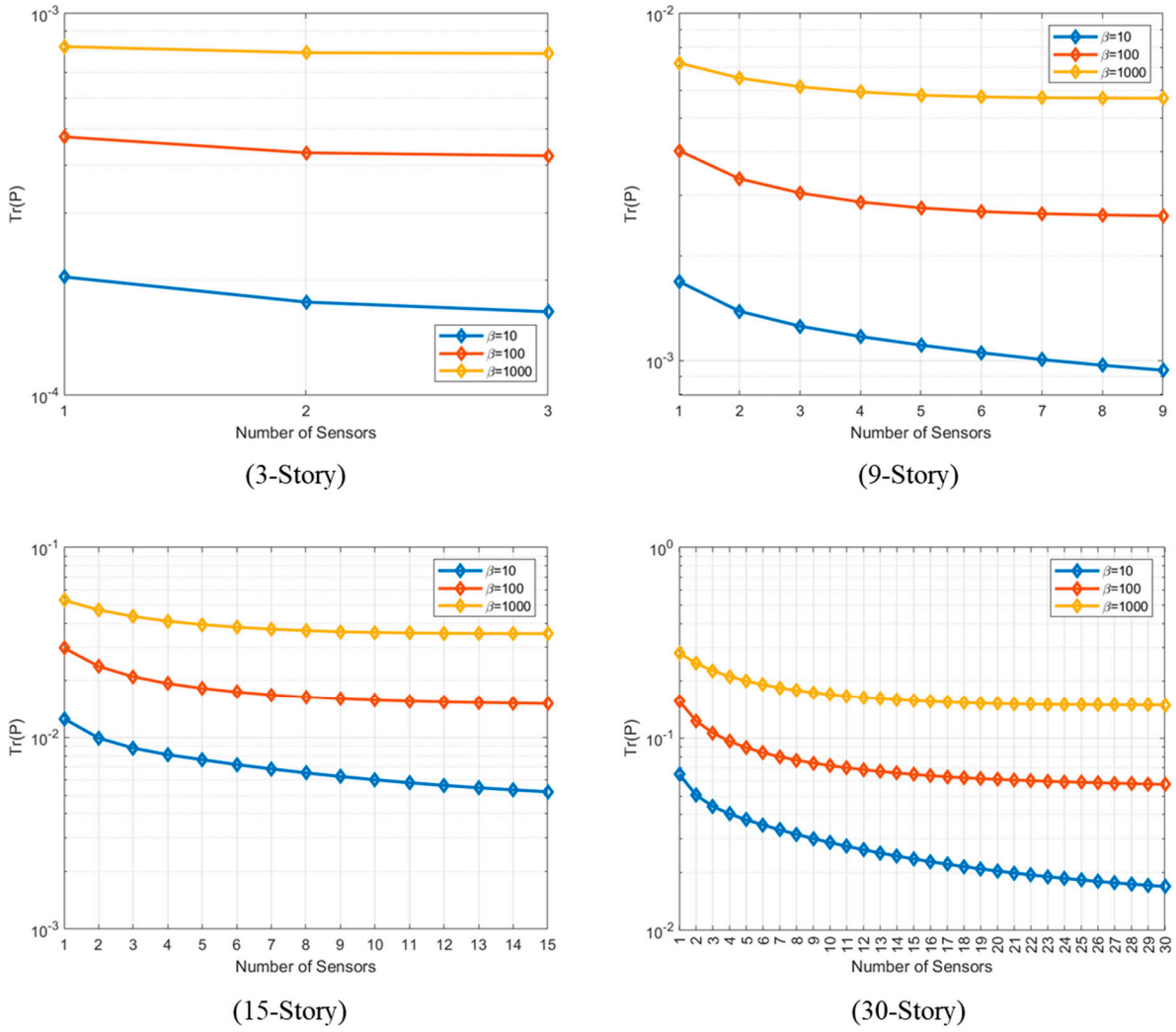


Figure 6. Pareto fronts for sensor placement optimization using the Kalman filter in 3-, 9-, 15-, and 30-story buildings, considering noise cases A (blue), B (red), and C (yellow).

In general terms, it can be observed that as the value of β increases, the curves shift upward along the vertical axis, indicating an increase in the second normalized objective, associated with the trace of the state estimation error covariance matrix $\mathrm{Tr}(P)$. This behavior reflects greater uncertainty in state estimation due to the increased noise in the measurements.

Furthermore, as the number of sensors increases (i.e., as f_1 grows), the value of the second objective decreases, highlighting the typical trade-off between estimation accuracy and instrumentation cost. In other words, the larger the number of sensors, the lower the uncertainty in state estimation.

It is also worth noting that the case of β = 1000, which represents an excessively high measurement noise level, lies in the upper region of the Pareto fronts. This area can be considered a non-feasible zone, as the associated estimation error becomes excessively large and impractical for structural health monitoring applications.

Tables 4 and 5 present the optimal sensor placement (OSP) configurations obtained for Cases (A) and (B), respectively. The OSP column indicates the selected floors in each con-

figuration, while the Tr(P) column shows the trace of the state estimation error covariance matrix, which is used to quantify the overall uncertainty of the estimated states.

Table 4. Optimized case sensor locations (A).

Building	Stories	OSP	Tr(*P*)
1	3	[3]	2.0×10^{-4}
2	9	[7,8,9]	1.3×10^{-3}
3	15	[1,12,13,14,15]	7.7×10^{-3}
4	30	[1,2,3,22,26,27,28,29,30]	3.0×10^{-2}

Table 5. Optimized case sensor locations (B).

Building	Stories	OSP	Tr(*P*)
1	3	[3]	4.8×10^{-4}
2	9	[7,8,9]	3.0×10^{-3}
3	15	[11,12,13,14,15]	1.8×10^{-2}
4	30	[23,24,25,26,27,28,29,30]	7.7×10^{-2}

In Case (A) (Table 4), it can be observed that as building height increases, a greater number of sensors is required to maintain acceptable levels of uncertainty. The value of Tr(P) increases from 2.0×10^{-4} in the 3-story building to 3.0×10^{-2} in the 30-story building, reflecting the growing difficulty of accurately estimating states in taller structures. Furthermore, the sensors tend to be more evenly distributed along the building height, occupying lower, middle, and upper levels. This distribution is highly effective for dynamic estimation: sensors at the top maximize its SNR by capturing the largest displacement amplitudes (dominating the fundamental mode), while sensors at the base provide necessary constraints to distinguish higher-order mode shapes, ensuring the filter can reconstruct the full dynamic response.

On the other hand, the general trend remains in Case (B) (Table 5); however, the increase in measurement noise requires a larger number of sensors, particularly in 15- and 30-story buildings. While Tr(P) values follow a similar pattern to Case (A), a deterioration in state estimation is observed. In this scenario, sensors are more frequently concentrated in the upper levels, suggesting a strategy aimed at capturing more significant structural responses in the presence of increased noise. This shift suggests that under high-noise conditions, the algorithm prioritizes the maximization of SNR by targeting the floors with the highest kinetic energy. While this concentration primarily targets the fundamental mode, the Kalman filter allows the system to infer the contribution of higher modes from these high-amplitude measurements, provided the structural model is sufficiently accurate and the system is stochastically observable.

In summary, the analysis shows that increasing measurement noise not only raises the uncertainty in state estimation (higher Tr(P) values) but also alters the sensor placement strategy, shifting sensors toward specific zones of the structure and requiring a greater number of devices to maintain reliable coverage.

4.4. Simulation Results 1: Normalized Root Mean Square Error (NRMSE)

The following presents the results of the simulation of sensor measurements, aimed at evaluating the Normalized Root Mean Square Error (NRMSE) in the state estimation of the building under study. The analysis focuses exclusively on the 9-story building, whose

detailed information is provided in Table 1. Additionally, the NRMSE values are compared across different sensor placement configurations, considering the 2010 Concepción ground motion record as a known perturbation.

This comparison allows assessing how each configuration influences the accuracy of the state estimation, where lower NRMSE values indicate better agreement between the estimated and actual structural responses.

4.4.1. Optimized Case with Different Levels of Measurement Noises

Table 6 shows the optimal sensor placement for the 9-story building obtained in Section 4.3. In this case, the accelerometers are located on floors 7, 8, and 9, forming the configuration that minimizes the multi-objective function defined in Equation (21).

Table 6. Optimized case sensor location for the 9-story building.

Case	Sensor Placement
Optimized	[7,8,9]

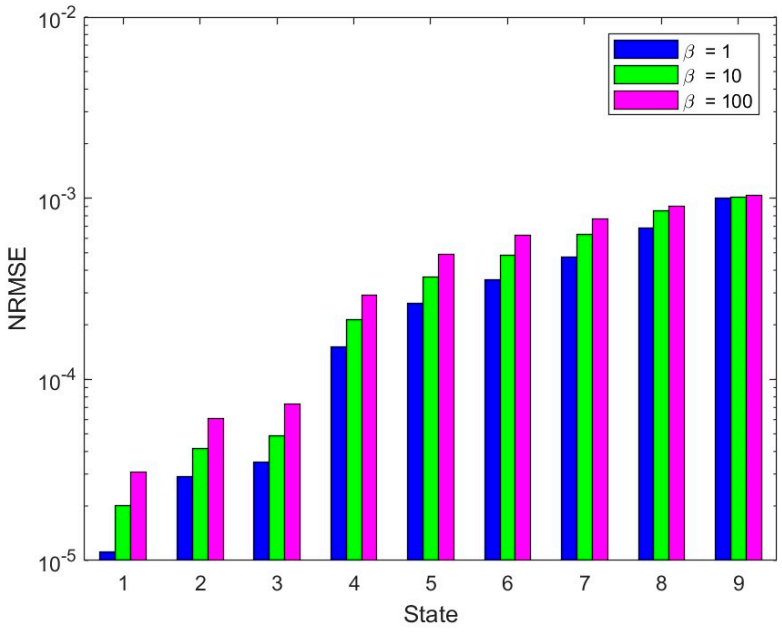
Table 7 presents the parameters corresponding to two scenarios with different levels of measurement noise, which are compared to a baseline scenario ($\beta = 1$) to analyze their impact on the percentage of error in state estimation. In this analysis, the value of Q_w is kept constant, while R_v progressively increases. Additionally, the table includes the SNR values (in dB), which reflect the signal-to-noise ratio associated with each configuration, and the resulting $\mathrm{Tr}(P)$ values, which represent the quality of the estimations. This approach allows evaluating how different levels of measurement noise influence the performance of the system.

Table 7. Parameters of three situations with different measurement noise levels.

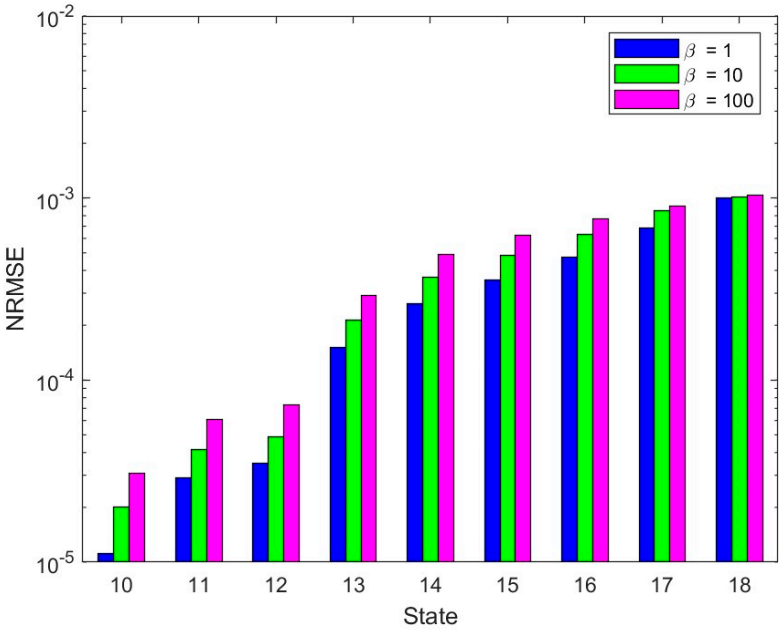
	Baseline	Case (A)	Case (B)
Q_w	0.001	0.001	0.001
β	1	10	100
R_v	0.001	0.01	0.1
SNR [dB]	38.3	28.3	18.4
$\mathrm{Tr}(P)$	7.0×10^{-4}	1.0×10^{-3}	3.0×10^{-3}

Figure 7 shows how the percentage of normalized absolute error in the estimation of modal displacements and velocities varies when using sensors with different levels of measurement noise, differentiated by their corresponding β values. These results are presented on a semi-logarithmic scale and confirm the hypothesis that, as measurement noise increases (higher values of R_v and β), the state estimation error also increases. This behavior is consistently observed in the graphs, where sensors with higher noise (β = 100) exhibit significantly larger errors compared to those with the baseline scenario (β = 1).

Furthermore, the relationship between measurement noise and estimation uncertainty is supported by the proportional increase in Tr(P) values, while the SNR values (in dB) show the same decreasing trend as R_v increases. These findings highlight the importance of selecting low-noise sensors to improve the accuracy of structural monitoring systems, even if this entails a higher associated cost that must be considered in the design.



(**a**) Modal displacement states



(**b**) Modal velocity states

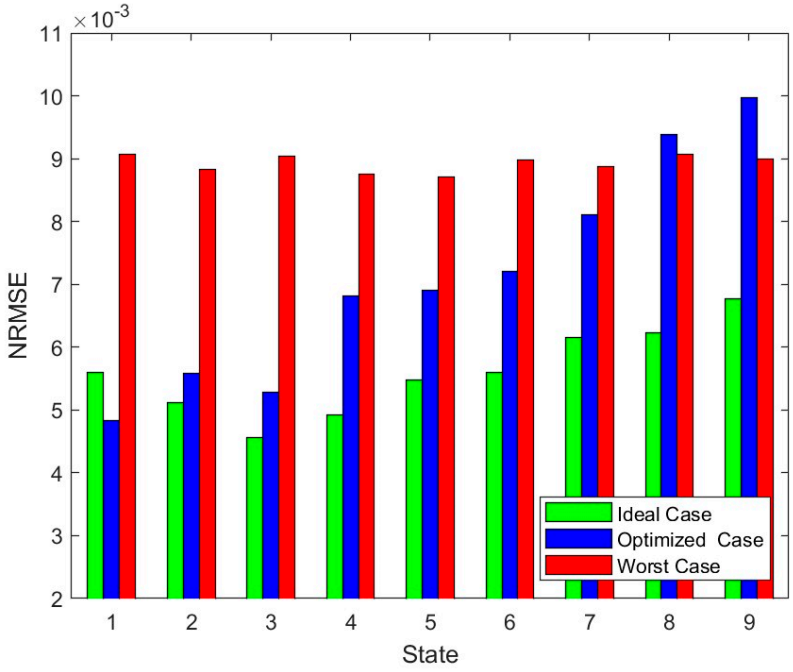
Figure 7. Comparison of normalized root mean square error (NRMSE) in state estimation for different noise levels (β = 1, 10, 100) under the excitation of the 2010 Concepción ground motion.

4.4.2. NRMSE Analysis of Ideal, Optimized, and Worst-Case Configurations

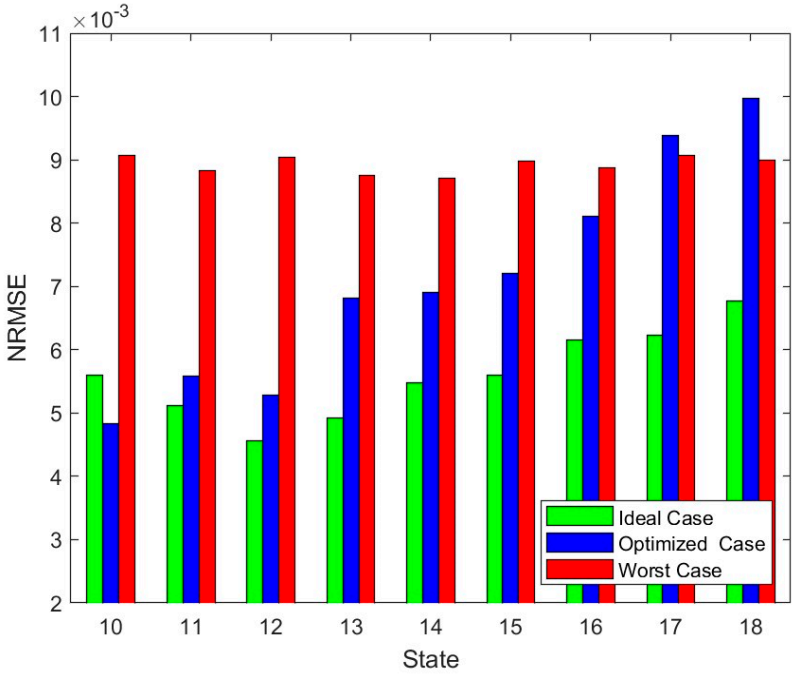
Table 8 presents a comparison between three sensor configurations for the 9-story building: the ideal, optimized, and worst-case scenarios. The comparison is based on the Normalized Root Mean Square Error (NRMSE) obtained from the state estimation results, which quantifies the accuracy of the reconstructed structural responses under each configuration.

The ideal case represents an extreme, costly, and impractical solution, in which sensors are located on every floor. The optimized case, in contrast, seeks an efficient balance between cost and estimation accuracy by strategically distributing the sensors to minimize the objective function. Finally, the worst-case configuration represents the minimum possible number of sensors (one), corresponding to the lowest estimation quality.

All configurations share the same noise parameters; however, the values of Tr(P) and SNR (dB) differ due to the specific sensor locations. The SNR is calculated by considering the sensor measuring the smallest signals, thus adopting a conservative approach to evaluate the system performance under the most challenging conditions. This comparison, visually illustrated in Figure 8, highlights how the number and placement of sensors directly affect the NRMSE values and, consequently, the overall quality of the state estimation. For this analysis, we consider the following parameter values: $Q_w = 0.001$ [m^2/s^4], $\beta = 10$, and $R_v = 0.01$ [m^2/s^4].



(**a**) Modal displacement states



(**b**) Modal velocity states

Figure 8. Comparison of NRMSE in state estimation for Ideal, Optimized, and Worst-Case sensor placement scenarios, under the excitation of a service-level earthquake modeled as white noise with PGA = 0.2 g.

Table 8. Location of sensors of the ideal case with the optimized case, and their respective Tr(P) values, under the excitation of a service-level earthquake modeled as white noise with PGA = 0.2 g.

Case	Sensor Placement	Tr(P)	Min SNR dB
Ideal	[9,8,7,6,5,4,3,2,1]	9.0×10^{-4}	15.9
Optimized	[9,8,7]	1.0×10^{-3}	16.4
Worst	[1]	8.0×10^{-3}	15.9

4.5. Simulation Results 2: Identification of Fundamental Frequencies

This section presents the results of the simulation of measurements obtained from sensors located in the optimal configuration determined using the Kalman filter-based approach. The objective is to evaluate the method's ability to identify the fundamental frequencies of the 9-story building through peak analysis in the Power Spectral Density (PSD). The study focuses exclusively on this building, as detailed information on its modal behavior is provided in Table 1.

Additionally, tables are included showing the percentage error associated with the identified fundamental frequencies compared to the theoretical values. Two types of excitations are considered: (i) a real ground motion recorded in the 2010 Maule earthquake, Concepción station; and (ii) a synthetic white noise ground motion. This analysis allows evaluating the accuracy of the proposed method and its ability to reliably estimate the dominant frequencies of the structure.

4.5.1. 2010 Concepción Ground Motion

Table 9 shows the identified frequencies with optimal sensor placement for the 9-story building. In this case, the accelerometers are strategically located on floors 7, 8, and 9. This configuration aims to maximize accuracy in the identification of the system's fundamental frequencies. For this analysis, the following simulation parameters were considered: $Q_w = 0.001 \left[\mathrm{m}^2/\mathrm{s}^4\right]$, $\beta = 10$ and $R_v = 0.01 \left[\mathrm{m}^2/\mathrm{s}^4\right]$.

Table 9. Comparison of theoretical fundamental frequencies with those estimated from the sensors in the optimized case, 2010 Concepción ground motion.

Mode	1	2	3	4	5	6	7	8	9
Fr. Original [Hz]	1.53	4.84	10.47	18.83	29.98	43.68	59.29	76.17	96.17
Fr. Found [Hz]	1.56	4.89	10.44	18.89	-	-	-	-	-
% Error	1.8%	1.1%	0.2%	0.3%	-	-	-	-	-

Figure 9 shows the power spectral density (PSD) curves obtained from simulated measurements by the sensors located according to the previously determined optimal configuration. In the graph, the dashed vertical lines indicate the theoretical fundamental frequencies of the building, while the PSD curves correspond to the measured coordinates of the sensors, represented on a semi-logarithmic scale. It can be observed that the peaks align with the theoretical lines, demonstrating that the measurements allow the identification of the first four fundamental frequencies of the structure with high accuracy. This result validates the effectiveness of the optimized sensor placement in capturing the main dynamic characteristics of the building under real seismic excitation.

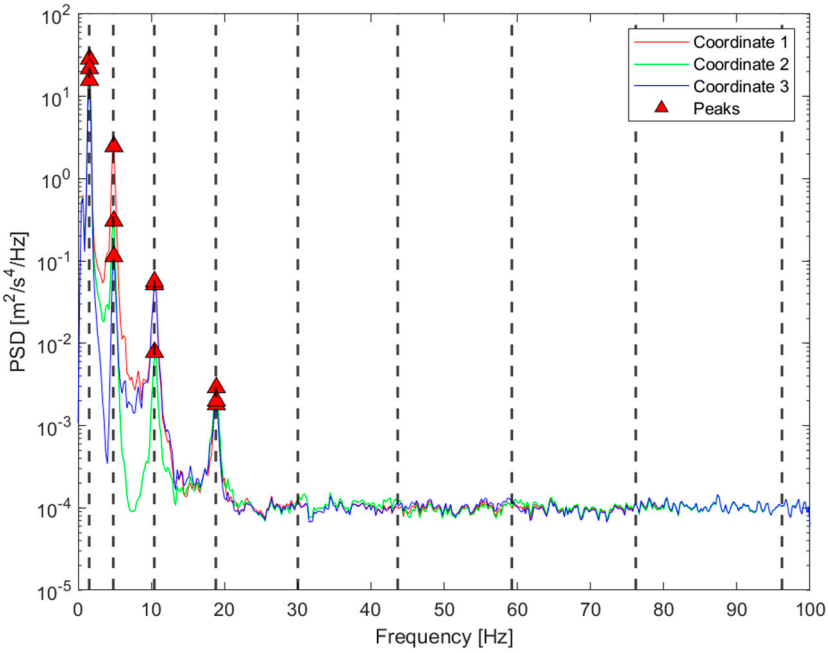


Figure 9. PSD curves of the sensor measurements of the optimized case, 2010 Concepción ground motion.

4.5.2. White Noise Ground Motion

To simultaneously excite all the relevant modes of the structure and evaluate the ability of the method to identify multiple fundamental frequencies, a band-limited white noise generated in MATLAB was used as the perturbation. The signal had a duration of 600 s, a sampling frequency of 1000 Hz, and a spectral content ranging from 0.00001 Hz to 100 Hz, defined by a 100th-order filter. This excitation was designed to fully cover the modal range of interest of the building.

Figure 10 shows the PSD curves obtained from simulated sensor measurements under white noise excitation using the same optimal sensor configuration for the 9-story building. As in the previous case, the dashed vertical lines represent the theoretical fundamental frequencies, while the peaks of the PSD curves correspond to the responses measured by the sensors. It can be observed that the identified peaks closely match the expected theoretical frequencies, indicating high accuracy of the proposed method in capturing the complete structural dynamics in this scenario.

Table 10 compares the theoretical fundamental frequencies of the building with those estimated from simulated sensor measurements, showing remarkable agreement between both sets of values. The relative estimation error, reported in the bottom row, remains very low, with a maximum of 1.8% and errors below 1.0% in most modes.

Table 10. Comparison of theoretical fundamental frequencies with those estimated from the sensors in the optimized case, white noise ground motion.

Mode	1	2	3	4	5	6	7	8	9
Fr. Original [Hz]	1.53	4.84	10.47	18.83	29.98	43.68	59.29	76.17	96.17
Fr. Found [Hz]	1.56	4.89	10.44	18.73	30.22	43.56	59.33	76.22	96.22
% Error	1.8%	1.1%	0.2%	0.5%	0.8%	1.3%	0.3%	0.1%	0.1%

These results demonstrate that, under the considered conditions, the optimized sensor configuration is capable of correctly identifying the nine fundamental frequencies of the building with very low error levels. In particular, the use of white noise as excitation

proves to be effective for exciting the different modes and validating the effectiveness of the proposed method.

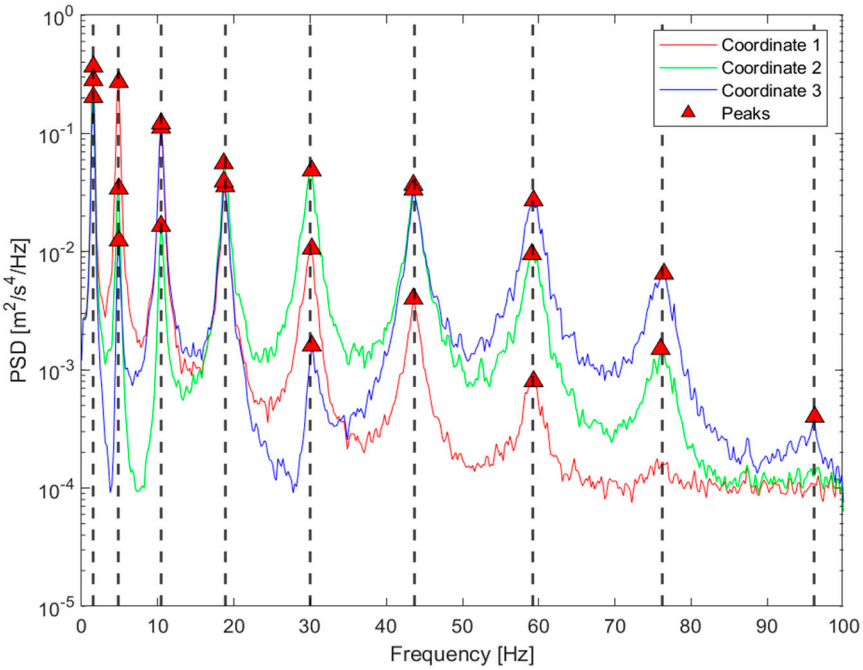


Figure 10. PSD curves of the sensor measurements of the optimized case, white noise ground motion.

5. Conclusions

In this study, a method based on the Kalman filter was developed and validated to optimize sensor placement in buildings, formulated as a multi-objective problem that simultaneously minimizes the number of sensors and the state estimation error. Using the simplified modeling approach of Miranda & Taghavi [27], together with the empirical equation of Guendelman et al. [28], representative models of 3-, 9-, 15-, and 30-story buildings were constructed, allowing the application and evaluation of the method across structures with various dynamic characteristics.

The trace of the state error covariance matrix ($\mathrm{Tr}(P)$) proved to be a representative performance criterion, directly correlating with both the signal-to-noise ratio (SNR) and the normalized absolute error in state estimation. This indicator enables the consistent quantification of uncertainty and effectively guides the search for optimal configurations, demonstrating that measurement noise is a critical factor in determining sensor layouts. As the noise covariance (R_v) increases, estimation uncertainty grows and placement strategies shift, requiring a greater number of sensors concentrated in specific high-energy structural zones. Conversely, high-sensitivity, low-noise sensors can significantly reduce uncertainty, confirming the fundamental balance between sensitivity and noise rejection required for robust monitoring systems.

Optimal sensor layouts tended to concentrate on upper floors where accelerations are larger, and the SNR is more favorable. It was also confirmed that maintaining a minimum SNR threshold of 10 dB allows for more reliable estimations, supporting the inclusion of this parameter as a primary design criterion. This strategy maximizes measurement quality without the need to instrument every floor, confirming that sensor redundancy at modal nodes or lower levels does not provide significant benefits to the global state estimation.

The proposed methodology demonstrated high accuracy in identifying the fundamental frequencies of structures under both synthetic white noise and real seismic perturbations, such as the 2010 Concepción ground motion record. These results validate the effectiveness

of the approach for structural health monitoring applications under realistic operating conditions. While this study establishes a robust theoretical framework, future research should transition to real-world applications by accounting for operational uncertainties, such as thermal variations, which can significantly influence material properties and structural response. Additionally, it is recommended to extend this framework to evaluate early damage detection capacity and explore hybrid sensor configurations—combining devices of varying sensitivity and cost—to enhance system robustness. Validating these strategies against open-source experimental benchmark datasets will be essential for demonstrating their practical applicability.

Author Contributions: Conceptualization, G.F.; methodology, G.F. and R.R.; software, R.R.; investigation, R.R.; resources, G.F.; writing—original draft preparation, R.R.; writing—review and editing, G.F. and R.R. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A. Modal Shapes of Buildings Under Study

The following figures present the modal shapes of the first 3 modes of vibration for the 3-, 9-, 15-, and 30-story building structures.

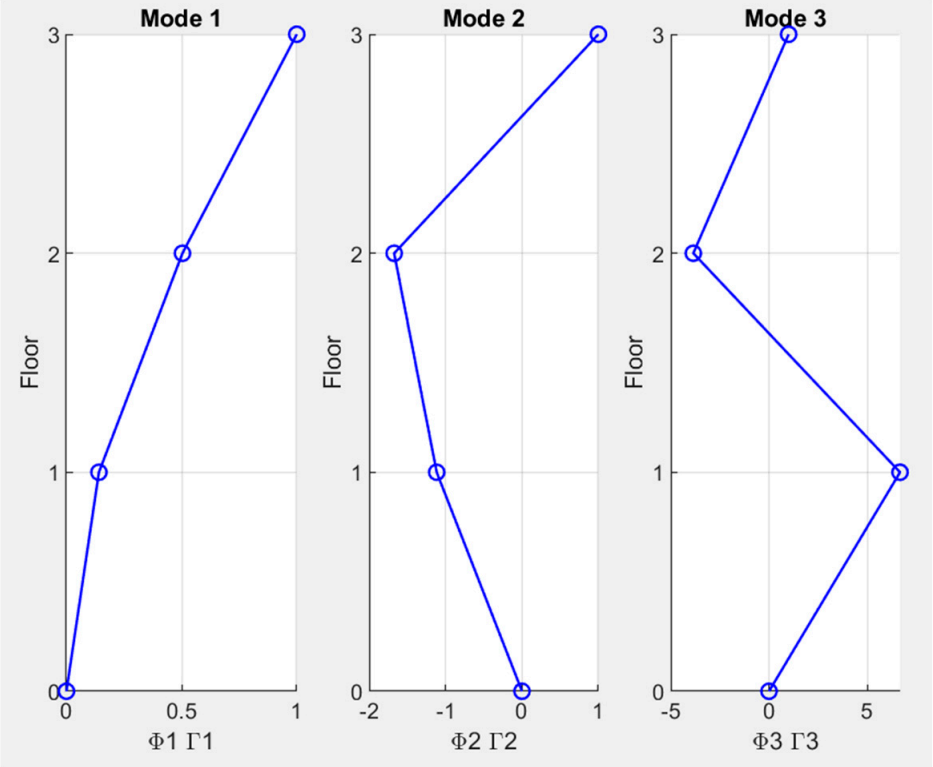


Figure A1. Modal shapes of the 3-story building.

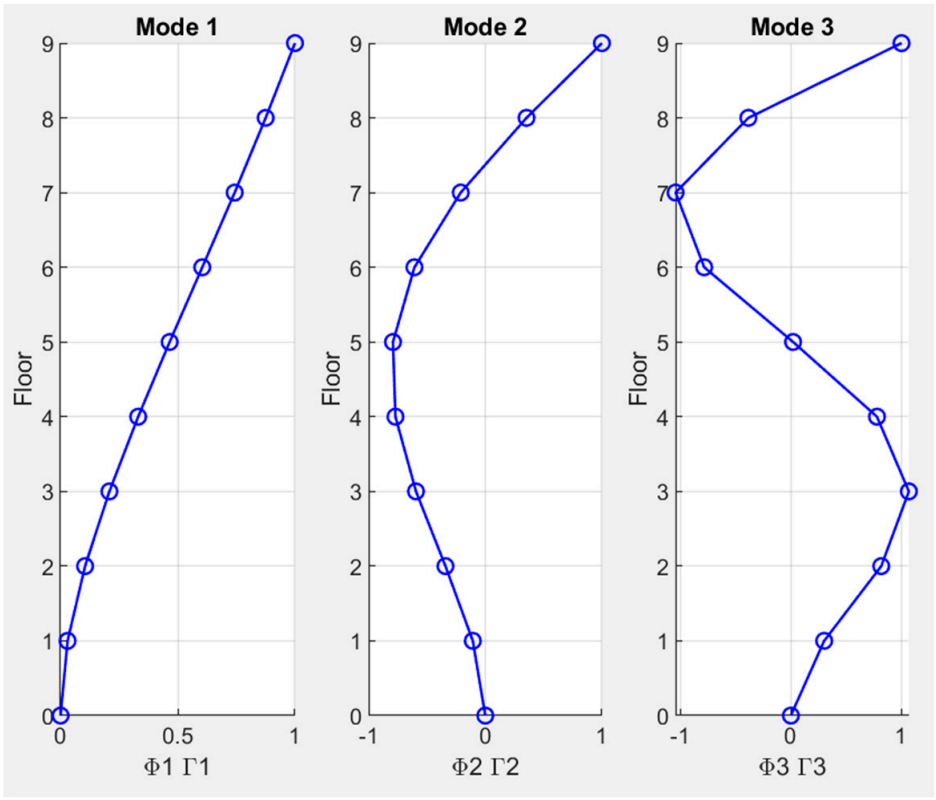


Figure A2. First three modal shapes of the 9-story building.

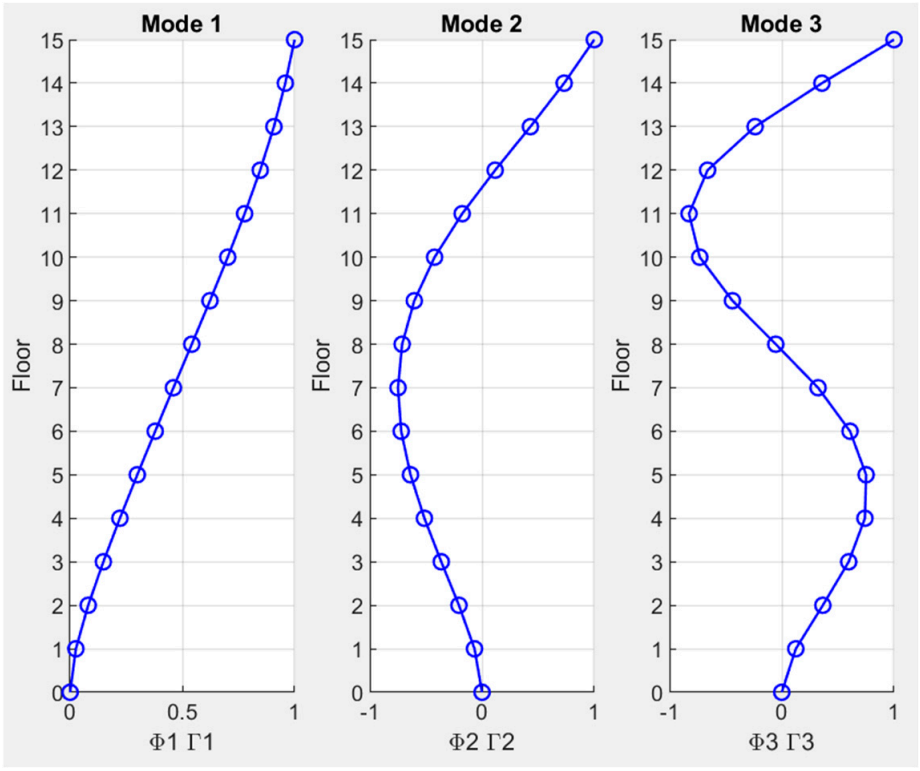


Figure A3. First three modal shapes of the 15-story building.

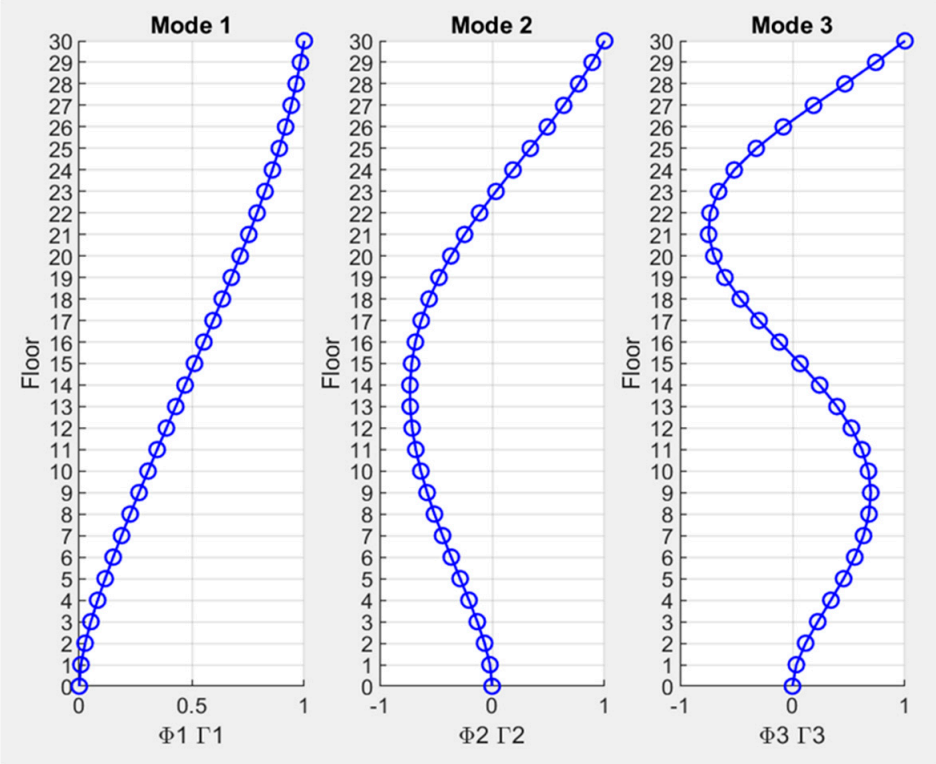


Figure A4. First three modal shapes of the 30-story building.

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